

Fall 1985

Problem 1 (Fa85). Evaluate the integral $\int_0^\infty \frac{1 - \cos ax}{x^2} dx$ for $a \in \mathbb{R}$.

Problem 2 (Fa85). Prove that for every $\lambda > 1$, the equation $ze^{\lambda-z} = 1$ has exactly one root in the disc $|z| < 1$ and that this root is real.

Problem 3 (Fa85). How many different monic irreducible polynomials of degree 2 are there over the field $\mathbb{Z}_5$? And of degree 3?

Problem 4 (Fa78, Fa85). Prove that every finite multiplicative group of complex numbers is cyclic.

Problem 5 (Fa85). How many zeros does the polynomial $z^4 + 3z^2 + z + 1$ have in the right half z -plane?

Problem 6 (Fa85). Let k be real, n an integer ≥ 2 , and let $A = (a_{ij})$ be the $n \times n$ matrix such that all diagonal entries $a_{ii} = k$, all entries $a_{i,i \pm 1}$ immediately above or below the diagonal equal 1, and all other entries equal 0. For example, if $n = 5$,

$$A = \begin{pmatrix} k & 1 & 0 & 0 & 0 \\ 1 & k & 1 & 0 & 0 \\ 0 & 1 & k & 1 & 0 \\ 0 & 0 & 1 & k & 1 \\ 0 & 0 & 0 & 1 & k \end{pmatrix}$$

Let $\lambda_{\min}$ and $\lambda_{\max}$ denote the smallest and largest eigenvalues of A , respectively. Show that $\lambda_{\min} \leq k - 1$ and $\lambda_{\max} \geq k + 1$.

Problem 7 (Fa85). Let $y(t)$ be a real valued solution, defined for $0 < t < \infty$, of the differential equation

$$\frac{dy}{dt} = e^{-y} - e^{-3y} + e^{-5y}$$

Show that $y(t) \rightarrow +\infty$ as $t \rightarrow +\infty$.

Problem 8 (Fa81, Fa85). Let f be a continuous function on $[0, 1]$. Evaluate the following limits.

$$\lim_{n \rightarrow \infty} \int_0^1 x^n f(x) dx$$

$$\lim_{n \rightarrow \infty} (n+1) \int_0^1 x^n f(x) dx$$

Problem 9 (Fa85). Let A be the symmetric matrix

$$\frac{1}{6} \begin{pmatrix} 13 & -5 & -2 \\ -5 & 13 & -2 \\ -2 & -2 & 10 \end{pmatrix}$$

As v ranges over the set of vectors for which $v^T A v = 1$, show that $\|v\|$ is bounded, and determine its least upper bound.

Problem 10 (Fa85). Let f and f_n , $n = 1, 2, \dots$, be functions from $\mathbb{R}$ to $\mathbb{R}$. Assume that $f_n(x_n) \rightarrow f(x)$ as $n \rightarrow \infty$ whenever $x_n \rightarrow x$. Show that f is continuous.

Problem 11 (Fa85). Let G be a subgroup of S_6 . Assume that G has an element of order 6. Prove that G has a normal subgroup H of index 2.

Problem 12 (Fa85, Sp20). Evaluate $\int_0^{2\pi} e^{e^{i\theta}} d\theta$.

Problem 13 (Fa85). Let $f(z)$ be analytic on the right half-plane $\mathbb{H} = \{z \mid \Re z > 0\}$ and suppose $|f(z)| \leq 1$ for $z \in \mathbb{H}$. Suppose also that $f(1) = 0$. What is the largest possible value of $|f'(1)|$?

Problem 14 (Fa85, Sp97, Fa98). Suppose that A and B are endomorphisms of a finite dimensional vector space V over a field $\mathbb{F}$. Prove or disprove the following statements:

1. Every eigenvector of AB is also an eigenvector of BA .
2. Every eigenvalue of AB is also an eigenvalue of BA .

Problem 15 (Fa85). Let $0 \leq a \leq 1$ be given. Determine all nonnegative continuous functions f on $[0, 1]$ which satisfy the following three conditions:

$$\int_0^1 f(x) dx = 1$$

$$\int_0^1 x f(x) dx = a$$

$$\int_0^1 x^2 f(x) dx = a^2$$

Problem 16 (Fa85). Let $f(x) = x^5 - 8x^3 + 9x - 3$ and $g(x) = x^4 - 5x^2 - 6x + 3$. Prove that there is an integer d such that the polynomials $f(x)$ and $g(x)$ have a common zero in the field $\mathbb{Q}(\sqrt{d})$. What is d ?

Problem 17 (Fa85, Sp98). Let (M, d) be a nonempty complete metric space. Let S map M into M , and write S^2 for $S \circ S$. Suppose that S^2 is a *strict contraction*; that is, there is a constant $\lambda < 1$ such that for $x, y \in M$, $d(S^2(x), S^2(y)) \leq \lambda d(x, y)$. Show that S has a unique fixed point in M .

Problem 18 (Fa85). Let G be a group. For any subset X of G , define its *centralizer* $Z(X)$ to be $\{y \in G \mid xy = yx, \text{ for all } x \in X\}$. Prove the following:

1. If $X \subset Y$, then $Z(Y) \subset Z(X)$.
2. $X \subset Z(Z(X))$.
3. $Z(X) = Z(Z(Z(X)))$.

Problem 19 (Fa85). An $n \times n$ real matrix T is *positive definite* if T is symmetric and $\langle Tx, x \rangle > 0$ for all nonzero vectors $x \in \mathbb{R}^n$, where $\langle u, v \rangle$ is the standard inner product. Suppose that A and B are two positive definite real matrices.

1. Show that there is a basis $\{v_1, v_2, \dots, v_n\}$ of $\mathbb{R}^n$ and real numbers $\lambda_1, \dots, \lambda_n$ such that, for $1 \leq i, j \leq n$:

$$\langle Av_i, v_j \rangle = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases} \quad \text{and} \quad \langle Bv_i, v_j \rangle = \begin{cases} \lambda_i & i = j \\ 0 & i \neq j \end{cases}$$

2. Deduce from Part 1 that there is an invertible real matrix U such that $U^T A U$ is the identity matrix and $U^T B U$ is diagonal.

Problem 20 (Fa85, Sp90). Let f be a differentiable function on $[0, 1]$ and let

$$\sup_{0 < x < 1} |f'(x)| = M < \infty$$

Let n be a positive integer. Prove that

$$\left| \sum_{j=0}^{n-1} \frac{f(j/n)}{n} - \int_0^1 f(x) dx \right| \leq \frac{M}{2n}$$